

DATA ENCODING STRATEGY

Part 1: Proof of Concept (CSV Data Preparation)

Project: MNC HR Attrition Prediction Model

Objective: Translate 32 raw HR attributes into mathematically precise formats for Machine Learning while preserving business logic and preventing data leakage.

1. TARGET VARIABLE

Column	Encoding Applied	Business Justification
Attrition	Binary Map (Yes=1, No=0)	Standard ML requirement. The model requires a numeric 1 or 0 to calculate loss and accuracy.

2. ORDINAL COLUMNS (Ranked Progression)

These columns contain text that represents a strict, measurable hierarchy. We must preserve the mathematical distance between levels.

Column	Encoding Applied	Business Justification
BusinessTravel	Ordinal Encoding	Explicit rank exists: Non-Travel (1) < Travel_Rarely (2) < Travel_Frequently (3).
Education	Ordinal Encoding	Explicit rank: Below College (1) < College (2) < Bachelor (3) < Master (4) < Doctor (5).
EnvironmentSatisfaction	Ordinal Encoding	Explicit rank: Low (1) < Medium (2) < High (3) < Very High (4).
JobInvolvement	Ordinal Encoding	Explicit rank: Low (1) < Medium (2) < High (3) < Very High (4).
JobLevel	Ordinal Encoding	Explicit rank: Entry (1) < Junior (2) < Mid (3) < Senior (4) < Executive (5).
JobSatisfaction	Ordinal Encoding	Explicit rank: Low (1) < Medium (2) < High (3) < Very High (4).
PerformanceRating	Ordinal Encoding	Explicit rank: Good (1) < Excellent (2) < Outstanding (3).
RelationshipSatisfaction	Ordinal Encoding	Explicit rank: Low (1) < Medium (2) < High (3) < Very High (4).
WorkLifeBalance	Ordinal Encoding	Explicit rank: Bad (1) < Low (2) < Good (3) < Better (4) < Best (5).

Note: Standard One-Hot Encoding is strictly REJECTED here because it would destroy the AI's ability to understand that "Very High" is mathematically greater than "Low".

3. NOMINAL COLUMNS: LOW CARDINALITY (Unordered Categories)

These columns have no mathematical relationship to one another (e.g., "Sales" is not greater than "HR"). Because they have 3 or fewer unique values, column explosion is not a risk.

Column	Encoding Applied	Business Justification
Department	One-Hot Encoding (OHE)	3 categories. Safely converts to 1s and 0s (e.g., Is_Sales, Is_HR) without bloating the dataset.
Gender	One-Hot Encoding (OHE)	2 categories. Converts to a single binary column (Is_Male 1 or 0).
MaritalStatus	One-Hot Encoding (OHE)	3 categories. Safely converts to binary indicators.
OverTime	One-Hot Encoding (OHE)	2 categories. Converts to a single binary column (Is_OverTime 1 or 0).

4. NOMINAL COLUMNS: HIGH CARDINALITY (Unordered Categories)

These columns have many unique text options (e.g., 9 unique Job Roles). Using OHE here would add 15 bloated, sparse columns of 1s and 0s, confusing the AI. We use advanced statistical encodings instead.

Column	Encoding Applied	Business Justification
JobRole (9 unique values)	Leave-One-Out Encoding	Replaces text (e.g., "Sales Exec") with the historical attrition % of that role. Calculates the mean excluding the current row to perfectly prevent the AI from "cheating" (data leakage) on our small CSV dataset.
EducationField (6 unique values)	Leave-One-Out Encoding	Compresses 6 categories into a single column representing the attrition probability of that degree, keeping the model lean and highly predictive.

5. EXCLUSIONS & NUMERIC DATA

- **Dropped Columns:** EmployeeCount, StandardHours, Over18.
Reason: Zero variance (constants). They provide no predictive power and waste compute.
- **Continuous Numeric Columns:** Age, MonthlyIncome, YearsAtCompany, etc.
Action: No encoding applied. These will undergo Min-Max Scaling (shrinking values to a 0.0-1.0 decimal scale) in a separate step so the AI can compare them fairly to the encoded categories above.